

EGA Controlled-Access Data

Step-by-Step Application Playbook

For Single-Cell Transcriptomic Reference Construction

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1. Overview

The European Genome-phenome Archive (EGA) hosts **879 single-cell studies** across 509 scRNA-seq, 231 10x Chromium, 58 spatial, 39 scATAC, 38 snRNA-seq, and other assay types. Unlike db-GaP's centralized Authorized Access, EGA uses a **per-study Data Access Committee (DAC)** model where each study's PI or institutional committee reviews access requests independently.

This playbook provides a **prioritized, step-by-step process** for efficiently applying to the highest-value DACs.

1.1. Key Constraint

We will only pursue EGA datasets for which the DAC permits **download to university on-premises HPC infrastructure**. Datasets requiring cloud-only processing (e.g., some UK Biobank datasets) are excluded from this phase.

2. Prerequisites

Before submitting any EGA applications:

1. **Create an EGA account:** <https://ega-archive.org/register/>
2. **Obtain institutional ethics approval:** Many DACs require an ethics committee opinion or IRB determination letter. Prepare a protocol synopsis describing the NMF reference construction project.
3. **Prepare a standard Data Transfer Agreement (DTA):** Some European DACs require a DTA or Data Processing Agreement between your institution and theirs. Have your institutional contracts office prepare a template.

4. **Install the EGA Download Client:**

```
pip install pyega3
```

Or the newer EGA Download Client v2:

```
pip install ega-download-client
```

5. **GDPR readiness:** Confirm your institution has Standard Contractual Clauses (SCCs) for EU-US data transfer, or can execute them on request.

3. Prioritized DAC Application Schedule

DACs are prioritized by the number of single-cell studies they cover. Apply to **Tier 1 first** (highest yield), then proceed to Tier 2 and Tier 3 as Tier 1 responses arrive.

3.1. Tier 1: Major Institutional DACs (Week 1–2)

DAC	Institution	~Studies	Focus
EGAC00001000205	Wellcome Sanger Institute	120	Human Cell Atlas, cancer, immune, developmental
EGAC00001000654	EMBL-EBI	40	Various European SC studies
EGAC00001000039	Karolinska Institutet	35	Developmental, immune, spatial
EGAC00001001093	Broad Institute	25	Cancer, immune — may overlap with dbGaP

Note

The Wellcome Sanger DAC alone covers ~120 SC studies. Getting this single approval unlocks the largest batch of EGA SC data. Prioritize this application above all others.

3.2. Tier 2: Specialized DACs (Week 2–4)

DAC	Institution	~Studies	Focus
EGAC00001000688	Max Planck Society	20	Developmental biology, aging
EGAC00001001077	University of Cambridge	18	Cancer immunogenomics, immune
EGAC00001000203	Helmholtz Munich	15	Metabolic disease, lung atlas
EGAC00001000789	Netherlands Cancer Inst.	12	Cancer multi-omics

3.3. Tier 3: Remaining DACs (Week 4+, rolling)

The remaining ~594 studies are distributed across ~100+ smaller DACs (often individual PIs). For these:

- Batch by disease category or tissue to allow template reuse.
- Contact DAC emails listed on each study's EGA page.
- Expect highly variable response times (1 week to 6+ months).
- Some PIs may never respond — deprioritize after two follow-ups.

4. Step-by-Step Application Process

4.1. Step 1: Identify the DAC for Each Study

Action: Find the DAC

For each target study (e.g., EGAS00001002072):

1. Visit <https://ega-archive.org/studies/EGAS00001002072>
2. Note the dataset IDs (EGAD...) listed under the study
3. Click each dataset to find the **Data Access Committee** field, which lists the DAC ID (EGAC...) and contact info
4. Record the DAC ID, contact email, and any institution-specific requirements

4.2. Step 2: Prepare the Application

Action: Customize the template

Use the `ega_dac_application_template.tex` file:

1. Replace [DAC NAME] with the specific DAC
2. Replace [DAC INSTITUTION] with their institution
3. Add the specific EGAS/EGAD IDs to the datasets section
4. Fill in your institutional details (university, IRB, etc.)
5. Compile to PDF: `pdflatex ega_dac_application_template.tex`

4.3. Step 3: Submit via EGA Portal

Action: Submit the request

1. Log in to <https://ega-archive.org>
2. Navigate to the dataset page (EGAD...)
3. Click “Request Access”
4. Fill in the online form with information from the PDF
5. Upload the PDF application as supporting documentation
6. Submit

Note

Some DACs have their own application forms hosted externally (e.g., the Sanger Institute uses a custom portal). Check the DAC page for institution-specific instructions before submitting.

4.4. Step 4: Follow Up

- If no response in 2 weeks: send a polite email to the DAC contact
- If no response in 4 weeks: send a second follow-up noting the original request date
- If no response in 8 weeks: consider the DAC non-responsive and deprioritize
- For institutional DACs (Sanger, EMBL, etc.): typical response is 2–4 weeks
- For PI-managed DACs: highly variable, often longer

4.5. Step 5: Download Approved Datasets

Action: Download to university HPC

Once approved, you will receive EGA download credentials:

```
# Configure pyega3
export EGA_USERNAME="your_ega_username"
export EGA_PASSWORD="your_ega_password"

# Download specific dataset
pyega3 -cf credentials.json fetch EGAD00001004149 \
  --output-dir /scratch/ega_data/

# Or use the newer client
ega-download-client download --dataset EGAD00001004149 \
  --output /scratch/ega_data/
```

4.6. Step 6: Confirm Compliance

After downloading:

- Verify data integrity (checksums)
- Confirm data is on encrypted storage
- Log the download in your data inventory
- Begin reprocessing pipeline
- Plan secure destruction after processing

5. Paired WGS + Single-Cell Studies in EGA

We identified **57 EGA studies** with paired WGS/WES and single-cell transcriptome data. These are **top priority** for genome–transcriptome linkage:

EGA ID	Study
EGAS00001002072	SC T cells from hepatocellular carcinoma patients
EGAS00001003387	SC RNA-seq + WGS from different cells
EGAS00001006807	Full-length SC RNA-seq with isoform + genomic alterations
EGAS00001002436	Subclonal evolution of ER+ breast cancers (WGS + scRNA)
EGAS00001006851	Multiregion WES + Smart-Seq3 scRNA of breast tumors
EGAS00001006052	Tumor cell mutations + antigen presentation (multi-omic)
EGAS00001005472	Glioma epigenetic encoding + transcriptional states
EGAS00001006374	Pathogenic variants + SC transcription in disease
EGAS00001007078	Clonally resolved SC multi-omics
	+ 48 additional studies (see catalog)

6. Protocol and Disease Distribution in EGA

Protocol		Breakdown:	
Protocol	Studies	Disease Category	Studies
scRNA-seq	509	Cancer	422
10x Chromium	231	Neurological	183
Spatial	58	Developmental	157
scATAC-seq	39	Normal/Atlas	148
snRNA-seq	38	Infectious	81
Smart-seq	31	Pulmonary	63
CITE-seq	19	Metabolic	30
Multiome	10	Kidney	21
snATAC-seq	8	Autoimmune	18
Drop-seq	1	Cardiovascular	14

Overlap with dbGaP/GEO: Only 0.2% of EGA studies mention GEO and 0.1% mention dbGaP, confirming that EGA represents a ~100% unique data layer.

7. Timeline and Tracking

Phase	Action	Target Date
Week 1–2	Submit Tier 1 (Sanger, EMBL, Karolinska, Broad)	$T_0 + 2$ wks
Week 2–4	Submit Tier 2 (MPI, Cambridge, Helmholtz, NKI)	$T_0 + 4$ wks
Week 4+	Rolling Tier 3 applications	$T_0 + 4$ –12 wks
Ongoing	Track responses, follow up, download approved data	Continuous

Maintain a shared spreadsheet tracking: DAC ID, submission date, dataset IDs, status (submitted / approved / denied / non-responsive), follow-up dates, download date, processing status.